

Lab8: Elections

Sophia Jia

3/8/2022

PART I: 2009 Iran Election

1. What is the unit of observation in the iran data frame? What are the dimensions?

The unit of observaiton in the iran data frame is the cated votes for 2009 Iran election in individual city. The dimensions is 336 x 9.

2. Which cities had the highest proportion of total votes cast for Ahmadinejad? Please return the top several city names along with the province name and the proportions.

```
iran %>%
mutate(ahmadinejad_votes_portion=ahmadinejad/total_votes_cast) %>%
arrange(desc(ahmadinejad_votes_portion)) %>%
  select(city,ahmadinejad_votes_portion)
```

```
## # A tibble: 366 x 2
##   city          ahmadinejad_votes_portion
##   <chr>                <dbl>
## 1 South Rodbar          0.978
## 2 Regan                  0.964
## 3 Qala Ganj             0.953
## 4 Anbarabad             0.943
## 5 Manojan               0.933
## 6 Zawa                  0.914
## 7 Ijrod                 0.905
## 8 Bawanat               0.889
## 9 Shadgan               0.881
## 10 Kamijan              0.877
## # ... with 356 more rows
```

South Rodbar and Regan have the highest proportion of total votes cast for Ahmadinejad.

3. Which cities had the highest proportion of total votes cast for Mousavi? Please return the top several city names along with the province name and the proportions.

```
iran %>%
mutate(mousavi_votes_portion=mousavi/total_votes_cast) %>%
arrange(desc(mousavi_votes_portion)) %>%
  select(city,mousavi_votes_portion)
```

```
## # A tibble: 366 x 2
##   city          mousavi_votes_portion
##   <chr>                <dbl>
## 1 Khaash              0.811
```

```
## 2 Sarawan 0.749
## 3 Chahbahar 0.718
## 4 Zabolli 0.700
## 5 Saibsooran 0.694
## 6 Bastak 0.694
## 7 Marawa Tapa 0.662
## 8 Kanarak 0.655
## 9 Mako 0.651
## 10 Nekshahr 0.634
## # ... with 356 more rows
```

Khaash and Sarawan have the highest proportion of total votes cast for Mousavi.

4. How many cities did Mousavi win?

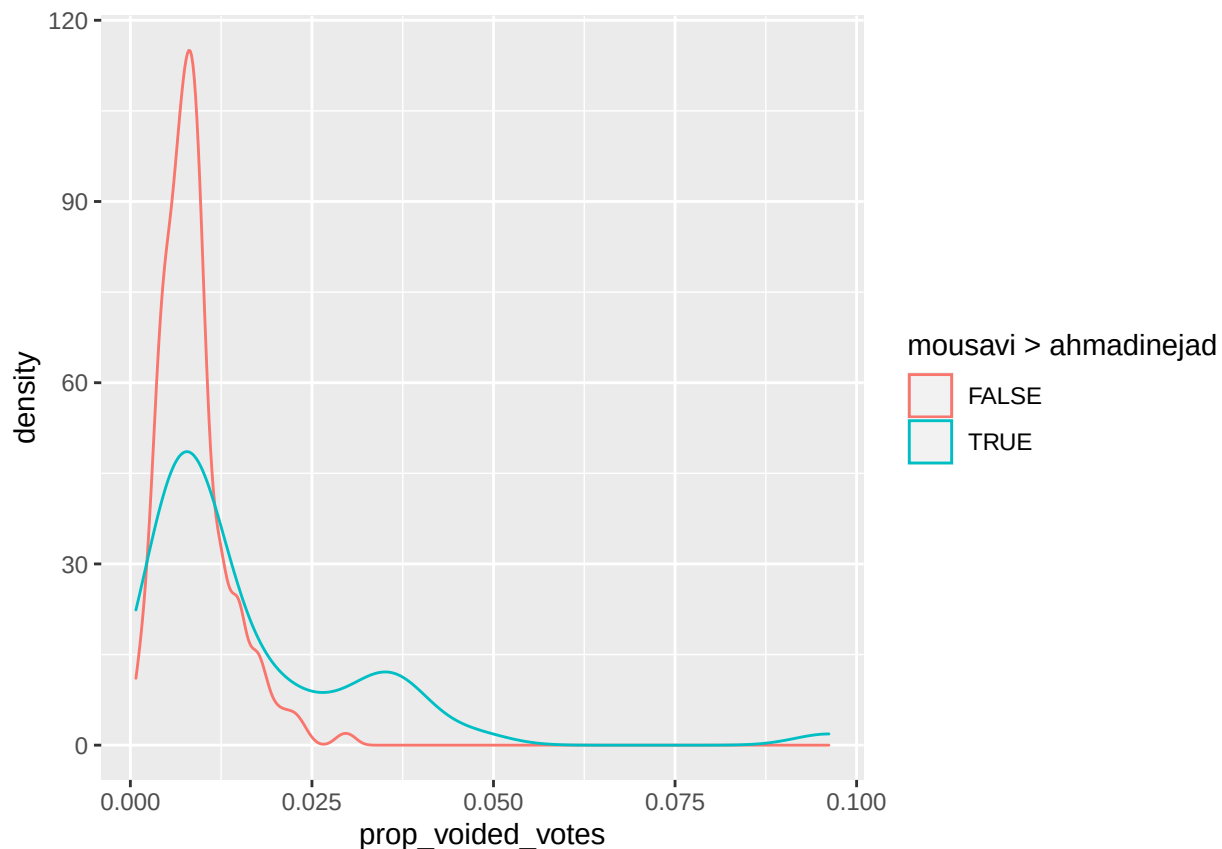
```
iran %>%
  group_by (ahmadinejad, mousavi) %>%
  mutate (mou_won = if_else (mousavi > ahmadinejad & mousavi > rezai & mousavi > karrubi, T, F)) %>%
  filter (mou_won == T) %>%
  nrow()
```

```
## [1] 46
```

There are 46 cities that Mousavi won.

5. How does the proportion of total votes that were voided compare between cities won by Mousavi and cities won by Ahmadinejad? This can be answered either with a plot or with summary statistics. Describe in words how they compare.

```
iran %>%
  mutate(prop_voided_votes = voided_votes/total_votes_cast) %>%
  filter ((mousavi>ahmadinejad&mousavi>rezai&mousavi>karrubi)|(ahmadinejad>mousavi&ahmadinejad>rezai&ahmadinejad>karrubi))
  ggplot(aes (x = prop_voided_votes, color = mousavi>ahmadinejad)) + geom_density()
```



It seems like Ahmadinejad has a lower voided votes than Mousavi, because Ahmadinejad has a higher intensity in the lower bound of the proportion voided votes.

6. What proportion of the total votes cast were won by Ahmadinejad and Mousavi, respectively?

```
iran%>%
  select (ahmadinejad,mousavi, total_votes_cast) %>%
  summarize (sum_ahm = sum (ahmadinejad),
             sum_mous = sum (mousavi),
             total_votes = sum (total_votes_cast)) %>%
  summarize (total_ahm = sum_ahm / total_votes,
             total_mous = sum_mous / total_votes)
```

```
## # A tibble: 1 x 2
##   total_ahm total_mous
##   <dbl>      <dbl>
## 1    0.625    0.337
```

Ahamadinejad's won total votes proportion is 62.47 % Mousavi's won total votes proportion is 33.70%

7. What proportion of vote counts for Ahmadinejad have "1" as a first digit?

```
iran %>%
  summarise (proportion = mean(get_first (ahmadinejad) == "1"))
```

```
## # A tibble: 1 x 1
##   proportion
```

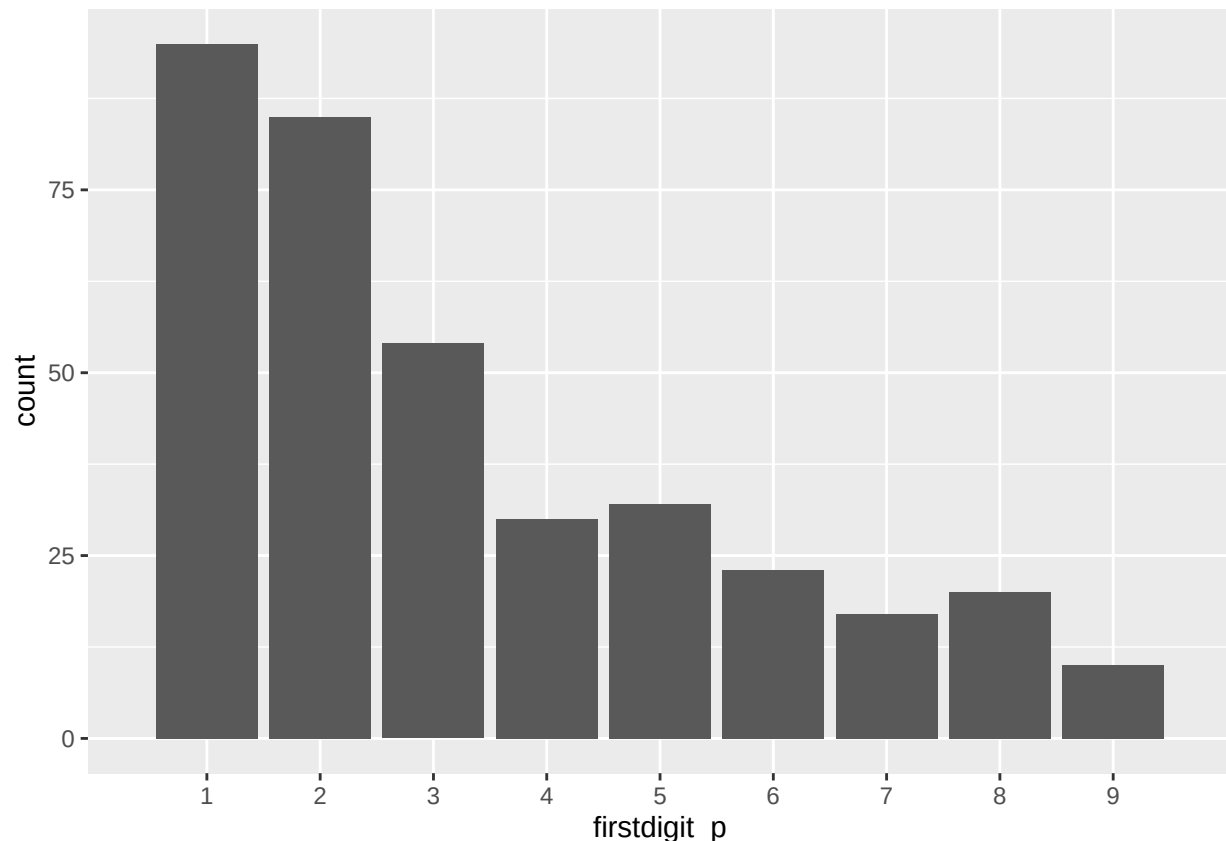
```
##      <dbl>
## 1      0.260
```

25.96% of votes counts for Ahmadinejad have “1” as a first digit.

8. Create a plot showing the distribution of first digits in the Ahmadinejad’s vote counts. Does this plot appear to match the ideal Benford’s distribution? Where does it deviate?

```
iran %>%
mutate (firstdigit_p = get_first (ahmadinejad)) %>%
ggplot(aes(x=firstdigit_p))+geom_bar() + scale_x_discrete(limits = 1:9)
```

```
## Warning: Continuous limits supplied to discrete scale.
## Did you mean 'limits = factor(...)' or 'scale*_continuous()'?
```



The data seems to be deviates in many places. Especially for digits stating with 2, 5, and 8, the proportion deviates in a large proportion. Benford’s law states that the count of the 1st digit decreases by half of the previous number each time. However, the graph above shows that the first digit proportion sometimes is not halved; rather, it actually increases in value.

9. Would you consider this meaningless, weak, moderate, or strong evidence of election fraud? Why or why not.

I think this is weak evidence of election fraud, because there’s only few instances that Benford’s law is followed. Especially at 2, 5, and 8, it increasees in proportion rather than decreasing by half as stated in Benford’s law. But more or less, the graph in question follows the same shape of the Benford’s Law. Too much elements such as political stand-point, environmental issue, should be considered; thus, this could only serve as a minimal evidence.

Part II: 2020 US Election

```
## Warning: One or more parsing issues, see 'problems()' for details
## Rows: 426582 Columns: 13
## -- Column specification -----
## Delimiter: ","
## chr (5): county, precinct, office, candidate, party
## dbl (6): district, votes, election_day, absentee, early_voting, affidavit
## lgl (2): machine_votes, absentee_hc
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

10. What state did you choose to study? What is the unit of observation in your state's data frame? What are the dimensions?

I choose New York to study. The unit of observation in New York's data frame is one precinct. The dimensions is 426582 x 13.

11. Create a table of the total votes won by each candidate for President, arranged in descending order. Who won in your state? Compare your table to a student that is studying a different state. Who won in their state? Which margin was larger?

```
newyork_votes %>%
  group_by(candidate) %>%
  summarise(sum = sum(votes, na.rm = TRUE)) %>%
  arrange(desc(sum))
```

```
## # A tibble: 653 x 2
##   candidate      sum
##   <chr>         <dbl>
## 1 Joseph R. Biden 5440643
## 2 Donald J. Trump 3420158
## 3 Ballots        913344
## 4 Blanks         524996
## 5 Under Votes    374418
## 6 Paul D. Tonko  346737
## 7 Blanks/Voids   308720
## 8 Carolyn B. Maloney 265172
## 9 Total          263408
## 10 Elizabeth L. Joy  240164
## # ... with 643 more rows
```

Joseph R. Biden won in the New York State. Comparing to another student who observed votes for West Virginia, Donald J. Trump won in their state. West Virginia (56.73%) has a larger margin than New York (37.14%).

12. Which 5 counties had the greatest number of total votes for Joseph Biden? Which 5 counties had the greatest number of total votes for Donald J. Trump? If there is a strong disagreement between those two lists, i.e., a county that cast lots of votes for Biden and few for Trump or vice-versa, do a bit of research into that county(s) and describe a bit about it (its principle cities, its economy, etc).

```
newyork_votes %>%
  filter(candidate == "Joseph R. Biden") %>%
```

```
group_by (county) %>%
  summarise(sum = sum(votes, na.rm = TRUE)) %>%
  arrange (desc(sum))
```

```
## # A tibble: 61 x 2
##   county      sum
##   <chr>      <dbl>
## 1 Kings      703310
## 2 New York   603040
## 3 Queens     569038
## 4 Nassau     396504
## 5 Suffolk    381021
## 6 Bronx      355374
## 7 Westchester 312437
## 8 Saratoga   273884
## 9 Erie       267174
## 10 Monroe    224691
## # ... with 51 more rows
```

```
newyork_votes %>%
  filter(candidate == "Donald J. Trump") %>%
  group_by (county) %>%
  summarise(sum = sum(votes, na.rm = TRUE)) %>%
  arrange (desc(sum))
```

```
## # A tibble: 61 x 2
##   county      sum
##   <chr>      <dbl>
## 1 Suffolk    381253
## 2 Nassau     326716
## 3 Saratoga   245220
## 4 Queens     212665
## 5 Kings      202772
## 6 Erie       197527
## 7 Monroe     145661
## 8 Westchester 144731
## 9 Richmond   123320
## 10 Onondaga   91715
## # ... with 51 more rows
```

It is interesting that New York County ranked in second place for Joe Biden but not even on the first ten for Donald Trump. New York City, Buffalo, and Rochester are the top 3 biggest city in New York. In total, New York carries 29 electoral votes based on its high population density. Especially for New York City, where Wall Street and Time Square are located; it is the economic center of the US and the world, containing lots of high-education workforce.

13. My Question: Which counties has the most voters in New York?

```
newyork_votes %>%
  group_by(county) %>%
  summarise(sum = sum(votes, na.rm = TRUE)) %>%
  arrange(desc(sum))
```

```
## # A tibble: 62 x 2
##   county      sum
```

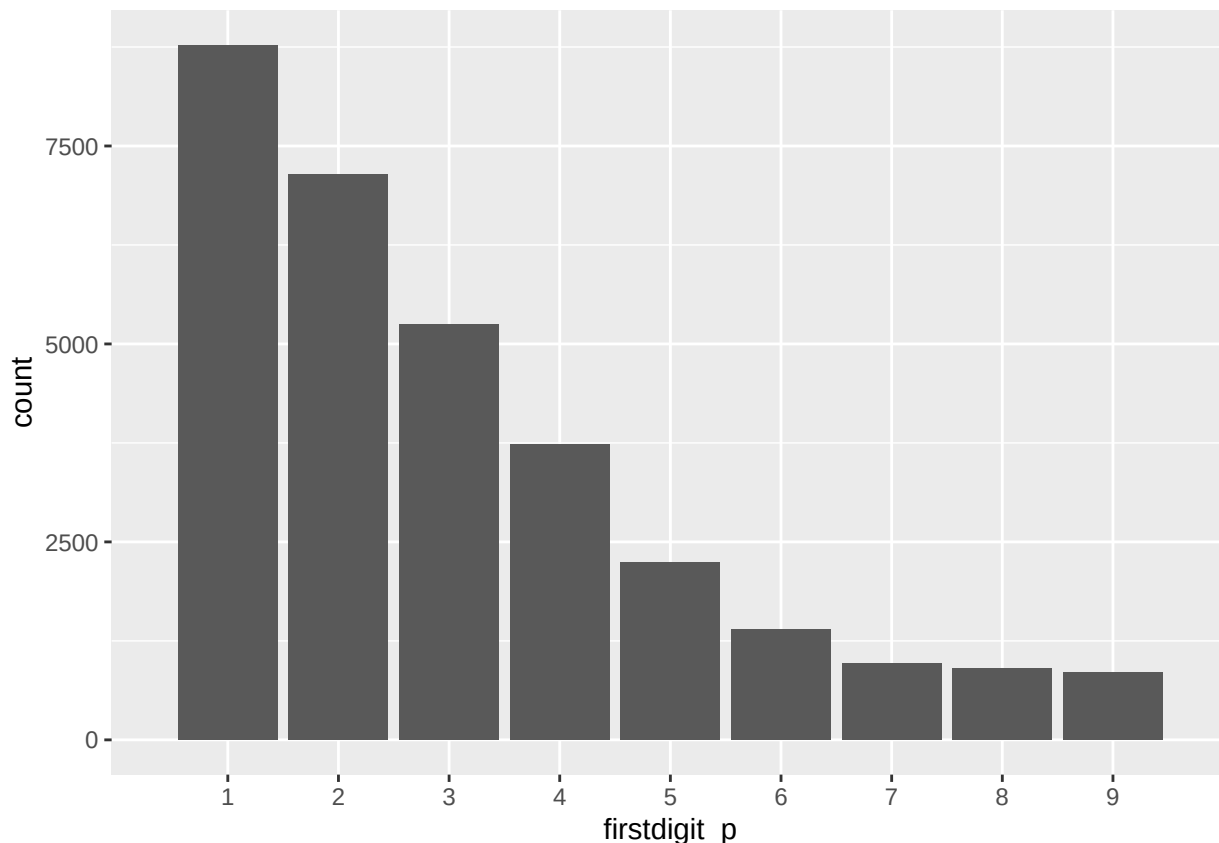
```
##      <chr>          <dbl>
## 1 Kings            3422155
## 2 Suffolk          3146204
## 3 Nassau           2948546
## 4 Queens           2862674
## 5 New York         2607780
## 6 Saratoga         2136896
## 7 Erie             1903507
## 8 Onondaga         1889339
## 9 Westchester     1875092
## 10 Bronx           1647344
## # ... with 52 more rows
```

Through this data table, I found that Kings has the most voters at 3422155 and Richmond has the least voters at 216153 voters.

14. Use this data to create a plot of that state's first digit distribution by precinct. Use the number of votes cast for Joseph Biden in each precinct.

```
newyork_votes %>%
  filter(candidate == "Joseph R. Biden") %>%
  filter (votes !=0) %>%
  group_by(precinct) %>%
  mutate (firstdigit_p = get_first (votes)) %>%
  ggplot(aes(x=firstdigit_p))+geom_bar(na.rm=TRUE) + scale_x_discrete(limits = 1:9)
```

```
## Warning: Continuous limits supplied to discrete scale.
## Did you mean 'limits = factor(...)' or 'scale*_continuous()'?
```



15. Does the election you chose appear to fit Benford's distribution better or worse than the Iran election?

It appears that the election data fit Benford's distribution better than the Iran election data. The count of the number is not nearly as equivalent as the number increases.

PART III

16. conduct a chi-squared goodness of fit test to determine whether or not the first digit counts from your state could have been drawn according to Benford's Law.

```
newyork_votes <- mutate(newyork_votes, digit = get_first(votes))
```

```
## Warning in get_first(votes): NAs introduced by coercion
```

```
ben_prop <- log10(1 + 1/1:9)
names(ben_prop) <- 1:9
ben_prop
```

```
##           1           2           3           4           5           6           7
## 0.30103000 0.17609126 0.12493874 0.09691001 0.07918125 0.06694679 0.05799195
##           8           9
## 0.05115252 0.04575749
```

```
obs_stat<- newyork_votes %>%
  mutate (digit = factor (digit)) %>%
  specify (response = digit) %>%
  hypothesize (null = "point", p = ben_prop) %>%
```

```

calculate (stat="Chisq")

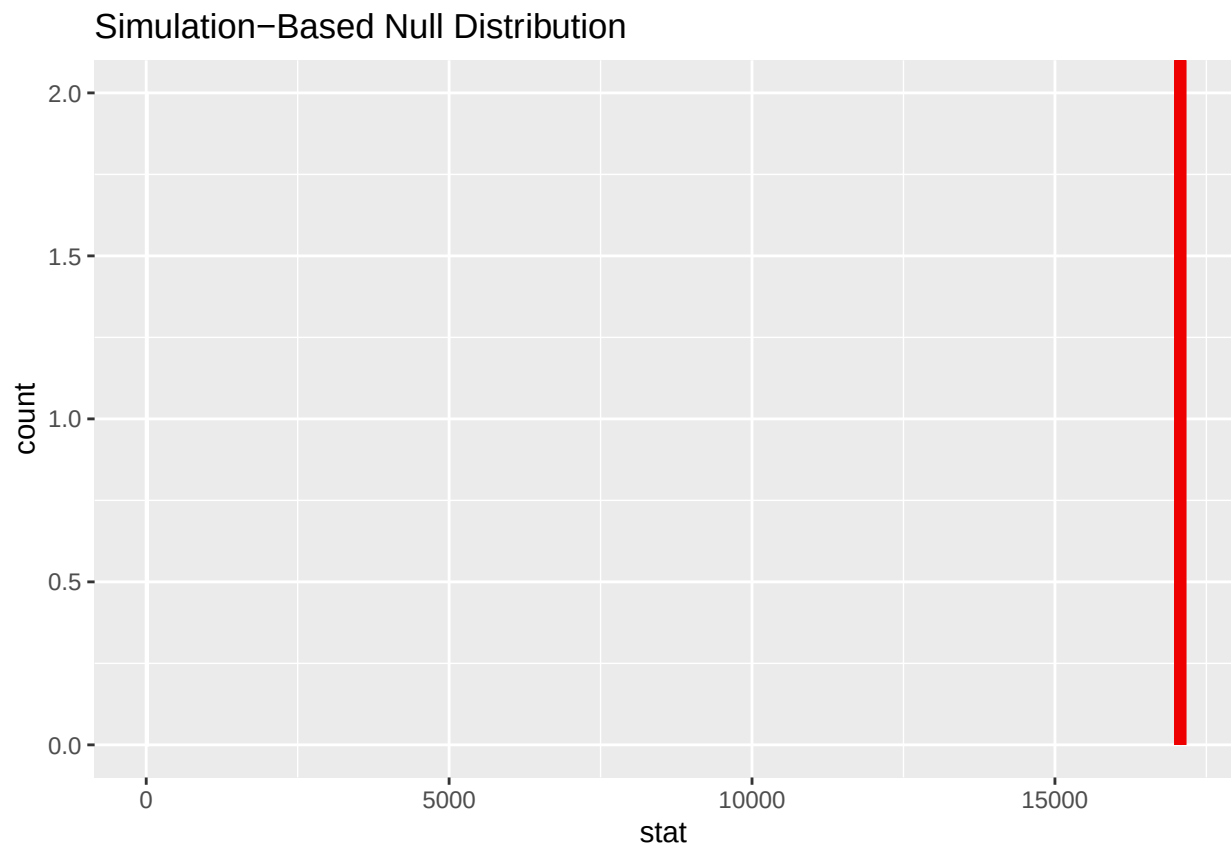
## Warning: Removed 838 rows containing missing values.
obs_stat

## Response: digit (factor)
## Null Hypothesis: point
## # A tibble: 1 x 1
##   stat
##   <dbl>
## 1 17071.

null<- newyork_votes %>%
  filter (votes !=0) %>%
  mutate(digit = factor(digit)) %>%
  specify(response = digit) %>%
  hypothesize(null = "point",
              p = ben_prop) %>%
  generate(reps = 10, type = "draw") %>%
  calculate(stat = "Chisq")

## Warning: Removed 337 rows containing missing values.
null %>%
  visualize() +
  shade_p_value(obs_stat = obs_stat,
               direction = "right")

```



```

null %>%
  get_p_value(obs_stat = obs_stat,
              direction = "right")

## Warning: Please be cautious in reporting a p-value of 0. This result is an
## approximation based on the number of 'reps' chosen in the 'generate()' step. See
## '?get_p_value()' for more information.

## # A tibble: 1 x 1
##   p_value
##   <dbl>
## 1      0

```

Null Hypothesize: First digit from the New York State follow the Benford's Law. Alternative Hypothesize: First digit from the New York State don't follow the Benford's Law.

I tried, but my computer only sustain reps to be 10, which is so minimal and does not generate significant results. P-value shows to be 0. At alpha level of 0.05, it means that we reject the null hypothesis. There's no systematic evidence that fair elections have vote counts that follow Benford's Law, so it's a flawed approach for detecting fraud.